

Shashank Mani Gautam

Indian Institute of Technology Kanpur, India

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WORK AND ACADEMIC EXPERIENCE

Project Engineer

[6 Sept 2025 – Present]

PI: Prof. Bishakh Bhattacharya, Indian Institute of Technology (IIT) Kanpur, Uttar Pradesh, India.

1. Design and Development of Skid-Steer mobile robot:

- Designed and implemented a dead-reckoning odometry pipeline utilizing wheel velocity measurements and IMU orientation data, providing motion estimates for map-based localization and navigation.
- Engineered a custom hardware interface for Maxon motors, enabling precise teleoperation via `ros2_control` and a gamepad controller
- Formulated and implemented LTV-LQR, MPC controller in gazebo world.
- Designed and integrated a graphical user interface (GUI) within the ROS2 ecosystem to stream and visualize real-time data from 3D LiDAR and thermal sensors.
- Implementing MPPI based mapping and autonomous navigation pipelines in real hardware.

2. Development of UGV-UAV for Inspection of Power substation :

- Developed the UGV-UAV architecture in ROS2 and Gazebo environment.
- Designed, simulated, and experimentally validating PID, LQR, LQG, and MPC-based guidance and control frameworks for autonomous precision landing of a PX4-based UAV on a moving UGV platform.
- Worked with Pixhawk 6c with PX4 and M9N GPS module and Implementation of the controller in holybro x500 drone.

3. Development of UAV using Visual Inertial Odometry:

- Developed the VIO in gazebo sim with UGV-UAV world with realsense camera.
- Utilized MSCKF algorithm with 20 clones to make the VIO worked in sparsed gazebo power substation environment.

Junior Research Fellow

[12 Aug 2024 – 27 Aug 2025]

PI: Dr. Ekta Singla, Indian Institute of Technology (IIT) Ropar, India

1. State of Art Design of manipulator:

- Designed and 3D printed a novel modular and reconfigurable robotic manipulator for mobile applications. [Video Link](#)

2. ROS2 Integration:

- Developed a modular ROS2-compatible library that auto-generates URDFs based on task-specific manipulator configurations.
- Implemented a custom ROS2 interface for real-time control of Dynamixel actuators.

- Performed the tele-operation of mobile modular reconfigurable manipulator using SSH communication and Raspberry Pi controller.
3. **Mobile robot for Medical Application:**
 - Developed a algorithm to drive the wheels with Raspberry Pi using ssh communication.
 - **Patent pending** for the system's novel design and reconfigurable functionality.
 4. **MATLAB Integration:**
 - Developed MATLAB scripts for communication and control of Dynamixel actuators.
 - Conducted workspace evaluation and dynamic analysis of the robotic manipulator system.

Teaching Assistant

[28 July 2023 – 20 May 2024]

Address: Indian Institute of Technology (IIT) Ropar, Rupnagar, Punjab 140001

- Assisted in teaching Engineering Drawing to undergraduate students.
- Conducted and demonstrated the Whirling of Shaft experiment in the Design Laboratory.

PATENTS AND PUBLICATION

1. **Gautam, Shashank Mani**, PATEL, Shreyas Maheshkumar, PADHEE, Srikant Sekhar, SINGLA, Ashish, SINGLA, Ekta. "RECONFIGURABLE MANIPULATOR ASSEMBLY" **Indian Patent No. [IN202511099317 A1]**, (2026).
2. **Gautam, Shashank Mani**, Ekta Singla and Ashish Singla. "Optimal Kinematic Synthesis and Prototype Development of Knee Exoskeleton" (2024).
3. **Shashank Mani Gautam**, Jefin T Jacob, Bishakh Bhattacharya, "Design and Development of Autonomous Skid Steer Unmanned Ground Vehicle integrable with UAV for substation health monitoring." Abstract submitted in ICoRD 26 Conference.
4. Anirudha Bhattacharjee, Nayan Jyoti Baishya, Pradeep Singh Chauhan, Subir Sen, **Shashank Mani Gautam**, Jefin T Jacob, Bishakh Bhattacharya, "AN INTELLIGENT AI ENABLED AUTONOMOUS MOBILE ROBOT FOR INSPECTION OF POWER SUBSTATION INFRASTRUCTURE" paper submitted in ASME IMEC26 Conference.

EDUCATION

Indian Institute of Technology (IIT) Ropar, Punjab, India

[26 Jul 2022 – 12 Jul 2024]

- M.Tech in Mechanics and Design (**Topper**), CGPA: 8.24/10.
- M.Tech Thesis: "Modelling, Design Optimization and Prototype Development of Lower Limb Knee Exoskeleton". [Link](#)
- Developed an optimized knee exoskeleton by using convex optimization in MATLAB.
- Published work link: <https://doi.org/10.48550/arXiv.2409.02635>.

National Institute of Technology, Kurukshetra, Haryana, India

[26 Jul 2017 – 12 Jul 2021]

- B.Tech Mechanical Engineering, CGPA: 8.60/10.
- Top 10% of the batch.

Central Hindu School, Gorakhpur, India

[2014 – 2016]

- Percentage in 12th (**Topper of the batch**): 89.9%
- CGPA in 10th (**Topper of the batch**): 10/10 CGPA

RESEARCH PROJECTS AND TRAINING

Lower Limb Knee Exoskeleton

[Jun 2023 – Jul 2024]

Research Scholar | PI: Dr. Ekta Singla, IIT Ropar

- Construct the optimal design algorithm for maximum range of motion of exoskeleton.
- Develop the CAD model based on optimal dimension and study of exoskeleton with mannequin model.
- My literature review allowed me to develop the algorithm for feasibility of exoskeleton with given optimal dimensions.
- The work helps to develop the Prototype of Knee exoskeleton based on linear actuator.
- Experiment Gait analysis of Sit to Stand motion of prototype.
- Misalignment study using Data analysis of gait of human knee joint and exoskeleton for sit to stand motion.

Beam Design

[Apr 2023 - May 2023]

Independent research project | PI: Dr. Rajendra Munian, IIT Ropar

- Created a discretization-based algorithm to solve beam deflection equations.
- Program predicted beam deflections with high accuracy compared to analytical results.

PUMA Robot Design

[Oct 2022 - Dec 2022]

Independent research project | PI: Dr. Manish Agrawal, IIT Ropar

- Designed and simulated a 4-DOF PUMA robot using Simscape-Simulink for torque and motion analysis.

Beam Analysis

[Jan 2020 - May 2020]

Independent research project | PI: Dr. Gian Bhushan, NIT Kurukshetra

- Simulated cantilever and simply supported beams under various loads to assess structural behavior.

Spline and Bezier curve

[Jan 2019 - May 2019]

Independent research project | PI: B V S Nagendra Reddy, NIT Kurukshetra

- Implemented MATLAB programs for smooth curve construction using spline (Ferguson method) and Bézier techniques.

CONFERENCE PARTICIPATION AND ORGANIZATION

- Participated in the workshop and Advance in Robotic 2025 conference conducted by IIT Jodhpur.
- Served as a **Coordinator** for the **Advances in Robotics 2023 International Conference** held at IIT Ropar, Punjab, India. [Link](#)
- Organizer in Conducting Workshop on Practicing robotics. Provide demo on reconfigurable and modular manipulator.

SCHOLARSHIPS AND AWARDS

- *ANSYS Scholarship* for scoring highest grade in my specialization | IIT Ropar | 2022.
- *Merit Scholarship* for standing first in the Senior High school | Central Hindu School | 2016.
- *Merit Scholarship* for standing first in the High school | Central Hindu School | 2014.
- *Science and Math Olympiad* for achieving first position (consecutive 4 years) | 2013-2016.

REFERENCE

Referee 1:

Prof. Bishakh Bhattacharya

Professor, Mechanical Engineering Department, IIT Kanpur, India

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Google Scholar ID: [scholar.com/ Bishakh Bhattacharya](https://scholar.com/Bishakh%20Bhattacharya)

Referee 2:

Dr. Ekta Singla

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Referee 3:

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